

# Harpreet Singh

## AI / Cybersecurity Researcher

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### Summary

AI and cybersecurity researcher with M.Sc. thesis work in uncertain-reasoning intrusion detection, DoS/DDoS detection, network traffic analysis, Bayesian Networks, Markov Networks, Zeek, Wireshark, Python ML, and applied data science. Broader engineering background in embedded systems, IoT, firmware, and FPGA acceleration supports security work on cyber-physical and edge-device systems.

### Core Skills

- Intrusion detection systems, DoS/DDoS detection, uncertain reasoning, Bayesian Networks, Markov Networks, probabilistic graphical models
- Zeek, Wireshark, TCP/IP, UDP, ICMP, MQTT, HTTP, WebSocket, SSL/TLS concepts, IoT security, embedded device security
- Python, TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, feature engineering, Grid Search, K-Fold Cross Validation
- Data engineering, data preprocessing, model evaluation, predictive analytics, NLP transformers, Punjabi BERT, medical AI prototypes
- Embedded systems, C, firmware, ARM Cortex, STM32, Ethernet/LWIP, sensors, edge AI, research publication and presentation

### Professional Experience

#### Research Assistant, Core-Hub Neuroengineering Solutions | *University of Lethbridge* | Apr 2025 - Present

- Build current PhD neuroengineering research stack for EEG/fNIRS acquisition, CGX/XDF dataset creation, live BCI classifier/game control, embedded device control, and FPGA temporal QiSNN acceleration.
- Developed an 8-channel EEG + 8-channel fNIRS Raspberry Pi platform using ADS1299, ADS8688, DAC8565, SPI/GPIO, TDM MUX sequencing, live plotting, and CSV logging.
- Built CGX/Sheeg workflows that launch CGX Acquisition and LabRecorder, record labelled visual-stimulus XDF sessions, preprocess EEG into CSV datasets, and export model artifacts for live classification.
- Developed closed-loop BCI/FPGA and rodent-decision pipelines using ADS1299 frames, MicroBlaze, BRAM windows, temporal QiSNN/SNN accelerators, UDP/UART intent packets, and five-state game control.

#### Research Assistant | *University of Regina* | Jan 2021 - Apr 2024

- Researched uncertain-reasoning IDS methods for DoS/DDoS detection using Bayesian Networks, Markov Networks, Zeek, Wireshark, feature engineering, and Python ML workflows.
- Published SECRYPT 2024 paper on uncertain-reasoning intrusion detection and validated IDS behavior through network traffic preprocessing, model tuning, and experimental evaluation.
- Built related ML prototypes including a Bayesian liver-disorder diagnostic model with 85% accuracy and Punjabi BERT/NLP classifiers for 100K+ news articles with 92% accuracy.

#### Independent Embedded Systems Developer | *Self-Employed* | Jun 2024 - Mar 2025

- Designed a custom STM32F405RG6 MicroPython development board, including multilayer PCB design, firmware flashing, sensor/actuator interfaces, power management, and hardware/software debugging.

#### Senior Software Engineer | *Planetcast Media Services Ltd.* | Dec 2018 - Nov 2020

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature acquisition platform.
- Designed schematics and PCBs, selected components, collaborated on cabinet/mechanical integration, and developed C firmware with Ethernet, HTTP/LWIP, alarm, and control interfaces.

#### Embedded Software Engineer / R&D Engineer | *Spark Eighteen Pvt. Ltd.; Exicom Tele-systems Ltd.* | Feb 2016 - Dec 2018

- Developed STM32/AVR embedded systems for LiFi control, telecom power plants, DC interface cards, and automatic test equipment.
- Designed 15-20 W isolated SMPS sections with 78-81% measured efficiency and performed board bring-up across CAN, SPI Flash, I2C EEPROM, Ethernet/LWIP, and GUI test automation.

### Selected Projects

#### Uncertain Reasoning IDS

- Developed DoS/DDoS intrusion detection models using Bayesian Networks, Markov Networks, feature engineering, Zeek/Wireshark traffic analysis, TensorFlow, PyTorch, and scikit-learn.

#### IoT / Embedded Security Background

- Combined embedded firmware, networking, secure-device awareness, Ethernet/MQTT/HTTP protocols, and traffic analysis experience for IoT and cyber-physical systems.

### **AI/ML Research Prototypes**

- Built probabilistic medical diagnosis and NLP classification systems, including a Bayesian liver-disorder model with 85% accuracy and Punjabi BERT classifiers with 92% accuracy.

### **Publication**

- Harpreet Singh et al., "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection," SECRIPT 2024, ISBN 978-989-758-709-2, ISSN 2184-7711, pp. 771-776.

### **Education**

- PhD in Computational Behavioural Neuroscience, University of Lethbridge - In progress
- M.Sc. Computer Science, University of Regina - 2024
- Bachelor's equivalent in Electronics & Telecommunication Engineering, AMIETE/IETE - 2012